



MODULE NAME:	MODULE CODE:
PRINCIPLES OF SECURITY	PRSE6212/w

ASSESSMENT TYPE: ASSIGNMENT 1 (PAPER ONLY)

TOTAL MARK ALLOCATION: 100 MARKS

TOTAL HOURS: 10 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Your assignment must be submitted through Turnitin.***
3. ***Save a copy of your assignment before submitting it.***
4. ***Assignments must be typed unless otherwise specified.***
5. ***All work must be adequately and correctly referenced.***
6. ***This is an individual assignment.***

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where **both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.

introductory phrases, accurate grammar, etc.	• An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

Question 1:**(Marks: 20)****Context:**

Significant threats to an enterprise can come from its own employees, contractors, and business partners, known as insiders, who pose an 'insider threat' due to their position as trusted individuals. Almost all businesses use external entities known as third parties or contractors. Examples of third parties include marketing agencies, landscapers, shredding contractors, and attorneys. Many businesses also use IT-related third parties due to their elevated level of expertise. For example, organizations often contract third parties to assist in developing software or apps (outsourced code development) or rely on third-party data storage facilities for managing critical data.

While third-party relationships can provide cost savings and operational efficiency, they also introduce various security risks or threats (insider threats), particularly when these third parties require access to an organization's computer network.

Question:

Critically evaluate the security risks associated with South African businesses that use third parties, particularly those that require access to the organization's computer network. In your response, address the following:

- Discuss how third-party system integration may expose businesses to security vulnerabilities, using the concept of "the weakest link." (8 marks)
- Provide examples of potential risks arising from third parties handling sensitive data or systems, such as those involved in outsourced code development or third-party data storage facilities. (6 marks)
- Analyze how the lack of expertise in system integration, combined with workarounds and incompatible systems, can create opportunities for cyber attackers. (6 marks)

Instructions:

Ensure that your response is well-structured, with clear examples, and provide in-text references where necessary. Your discussion should demonstrate a deep understanding of third-party security risks and practical ways to manage and mitigate these risks.

Question 2**(Marks: 20)****Authentication and Password Security in the South African Banking Industry (20 Marks)****Background:**

Passwords have long been the most common form of authentication, especially in the banking sector. However, with the growing sophistication of cyberattacks, such as brute force attacks, dictionary attacks, and password spraying, passwords alone are no longer considered sufficient for protecting sensitive financial information. The banking industry in South Africa faces constant pressure to ensure secure authentication methods to prevent unauthorized access to customer accounts.

The weakness of passwords lies in the human memory and behavior: people often create weak, predictable passwords or reuse the same passwords across multiple accounts. These behaviors increase the vulnerability to various password attacks. As artificial intelligence (AI) continues to advance, both attackers and defenders in the cybersecurity landscape are beginning to rely on AI. In banking, AI can be a game-changer for both password attacks and the defense mechanisms against them.

Q.2.1	Explain the following password attack methods and how they may be used by cybercriminals in the South African banking industry: • Brute Force Attack • Dictionary Attack • Password Spraying • Pass the Hash Attack • Hybrid Attack	(10)
Q.2.2	Research the emergence of AI in cybersecurity, focusing on its implications for passwords as a type of authentication in the South African banking industry. Your research should address the following: • The role AI plays in strengthening password protection. • How AI can be used by attackers to improve password-cracking methods. • Alternative authentication methods that AI supports (e.g., biometrics, behavioural authentication) and whether these could replace passwords in the banking sector. • Include examples of South African banks or international financial institutions that have adopted AI-driven security measures.	(10)

Question 3**(Marks: 30)****Firewall Recommendation Report for a South African SMME****Background:**

Thando Mthembu is the owner of **The African Taste Co.**, a small business in South Africa that specializes in exporting locally produced spices and sauces. The business employs 50 full-time staff and has an annual turnover of approximately R10 million. While business is booming, Thando is concerned about the security of his growing company's network and customer data as they expand their online operations and open a new e-commerce platform.

Thando has recently heard about different types of firewalls but lacks the technical knowledge to make an informed decision about which firewall is most appropriate for **The African Taste Co.** He has heard about several firewall categories:

1. stateful vs. stateless
2. open source vs. proprietary
3. hardware vs. software and
4. host-based vs. virtual firewalls

Thando, however, is unsure of which solution would best meet the needs of his business and maintain the integrity of his company's IT infrastructure.

Task:

Thando has asked you, as a network security consultant, to write a report recommending the most suitable firewall for **The African Taste Co.** The report must consider the firewall categories mentioned above and how they apply to his business.

In your report:

- a. Compare the various categories of firewalls: stateful vs. stateless, open source vs. proprietary, hardware vs. software, and host-based vs. virtual.
- b. Recommend the most appropriate type of firewall for **The African Taste Co.** based on its size, industry, and network security needs.
- c. Provide reasoning for your recommendation by explaining the pros and cons of each firewall type within the context of Thando's business.

- d. Ensure that your recommendation is **based on real-world products** and not on general categories of firewalls.
- e. Use credible sources to support your recommendation and provide references for your research.

Mark Distribution for Report on Recommended Firewall Solution and Report Structure:

1. Executive Summary (4 marks)

- Clarity and conciseness of the summary (2 marks)
- Relevance to the business context (2 marks)

2. Background (4 marks)

- Understanding of the business (2 marks)
- Identification of the need for a firewall (2 marks)

3. Firewall Categories Considered (8 marks)

- Comprehensive explanation of each category (4 marks)
- Comparison of advantages and disadvantages (4 marks)

4. Recommendation (10 marks)

- Clear recommendation of the firewall solution (2 marks)
- Justification of the recommendation based on business needs (4 marks)
- Consideration of cost and technical requirements (4 marks)

5. Conclusion (2 marks)

- Summarization of key points (1 mark)
- Final statement on the importance of the chosen solution (1 mark)

6. References (2 marks)

- Use of credible sources (1 mark)
- Proper citation format (1 mark)

Question 4**(Marks: 30)****Background:**

In South Africa, the widespread security threat landscape presents significant challenges for home users, including the risks of theft, house break-ins, and car break-ins. Mobile devices such as laptops and tablets are particularly vulnerable, often containing sensitive personal information and data. With the increasing reliance on technology in daily life, home users must consider effective security measures to protect their digital assets.

One potential solution is the implementation of a Trusted Platform Module (TPM) in their devices.

Question:

Discuss how the Trusted Platform Module (TPM) can be valuable to South African home users in enhancing the security of their mobile devices against prevalent security threats. Your discussion should cover the following points:

1. An overview of what TPM is and its key functionalities.
2. The specific benefits of TPM in the context of South African home users.
3. How TPM can mitigate risks associated with theft and unauthorized access to sensitive data.
4. Recommendations for home users on implementing TPM in their devices.
5. A consideration of any limitations or challenges associated with using TPM.

Mark Distribution and suggested structure of your answer:

1. **Overview of TPM and Key Functionalities (5 marks)**
 - Clear explanation of what TPM is, including its role and key functionalities (e.g., cryptographic services, key generation, system integrity checks).
2. **Specific Benefits of TPM for South African Home Users (7 marks)**
 - Discussion of how TPM specifically benefits South African home users in terms of security.
 - Mention of the relevance of TPM in addressing local security concerns.
3. **Mitigating Risks from Theft and Unauthorized Access (8 marks)**
 - Explanation of how TPM helps mitigate risks related to theft (e.g., protecting sensitive data on stolen devices).
 - Discussion of mechanisms TPM employs to prevent unauthorized access.

4. Recommendations for Home Users on Implementing TPM (5 marks)

- Practical suggestions for home users on how to implement TPM in their devices.
- Consideration of user-friendliness and ease of adoption.

5. Limitations and Challenges of Using TPM (5 marks)

- Discussion of any limitations or challenges associated with TPM, such as compatibility issues or user awareness.
- Consideration of potential drawbacks or barriers to effective implementation.

Note: Make sure your work is evidence-based by citing all references used.

END OF PAPER